TESTs - Lesson 11 – Tests - TokenSale.sol

Why did the test initially fail when trying to read `total_supply` from `tokenContract`?

- A The network was down.
- B The contract did not have a `total_supply` function implemented.
- C The contract was deployed with a wrong address, making it unrecognized as a valid ERC20 token.
- D The `total_supply` function was misspelled.

What is the name of the token contract that `TokenSale` will use for payments?

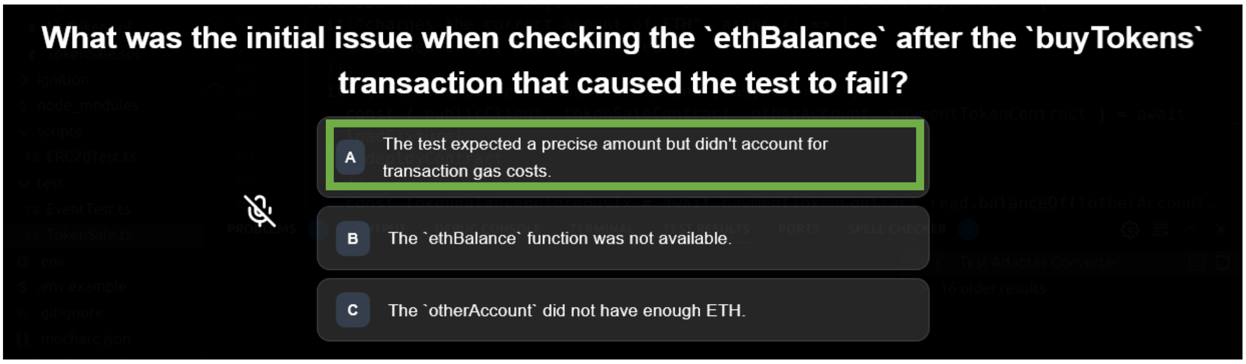
- A MyNFT
- B NFTShop
- C MyToken
- D TokenSale

What are the two key modifiers suggested for the `buyTokens` function definition to enable it to receive payments?

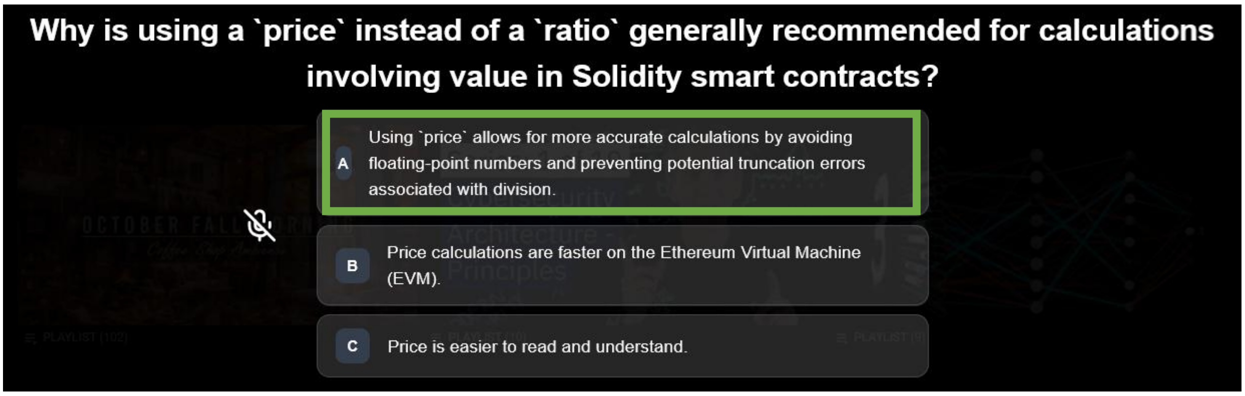
- A public and payable
- B internal and constant
- C private and view

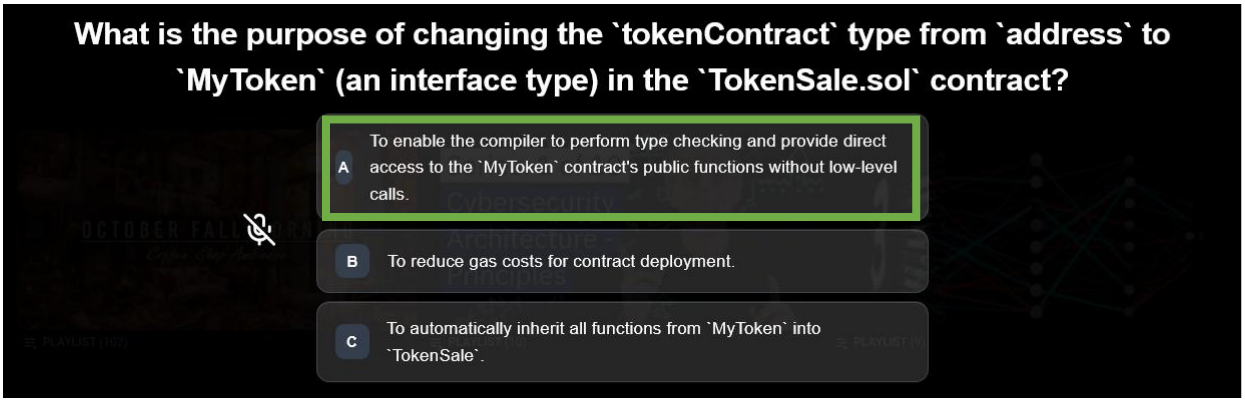
How is the `amountToBeMinted` calculated in the `buyTokens` function?

- A msg.value * ratio
- B msg.sender * ratio
- C msg.value * price
- D msg.value + ratio


What was the initial issue when checking the `ethBalance` after the `buyTokens` transaction that caused the test to fail?
A The test expected a precise amount but didn't account for transaction gas costs.
B The `ethBalance` function was not available.
C The `otherAccount` did not have enough ETH.


What is the primary difference between handling native tokens (like ETH) and ERC20 tokens for payments within a Solidity smart contract, as demonstrated in the video?
A Native tokens require explicit `transfer` calls, while ERC20s are handled automatically by the `payable` keyword.
B ERC20 tokens require prior `approve` and subsequent `transferFrom` or `burnFrom` calls, whereas native tokens are handled directly with the `payable` keyword.
C Native tokens use `msg.value`, while ERC20 tokens use `msg.sender`.


Why is using a `price` instead of a `ratio` generally recommended for calculations involving value in Solidity smart contracts?
A Using `price` allows for more accurate calculations by avoiding floating-point numbers and preventing potential truncation errors associated with division.
B Price calculations are faster on the Ethereum Virtual Machine (EVM).
C Price is easier to read and understand.


What is the purpose of changing the `tokenContract` type from `address` to `MyToken` (an interface type) in the `TokenSale.sol` contract?
A To enable the compiler to perform type checking and provide direct access to the `MyToken` contract's public functions without low-level calls.
B To reduce gas costs for contract deployment.
C To automatically inherit all functions from `MyToken` into `TokenSale`.

Which of the following best describes the core functionality of the `buyTokens` function in `TokenSale.sol` after all implementations in the video?

- A. It enables users to purchase ERC20 tokens by sending ETH to the contract, minting new tokens based on a fixed ratio.
- B. It facilitates the burning of ERC20 tokens to receive ETH back.
- C. It updates the ratio of ERC20 tokens to ETH based on market demand.
- D. It allows users to mint new ERC721 NFTs by paying ETH.

What common OpenZeppelin ERC20 extension was imported and utilized in `MyToken.sol` to allow for the removal of tokens from circulation?

- A. ERC20Pausable
- B. ERC20Permit
- C. ERC20Snapshot
- D. ERC20Burnable